

Operations Playbook for Local RAG Indexes

Chapter 01 - Ingestion Contract

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 001 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 101 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 02 - Chunk Metadata

This pdf-playbook note describes every chunk carries source, title, section, chunk_id, offsets, and page information when available. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 002 repeats the phrase chunk metadata to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes every chunk carries source, title, section, chunk_id, offsets, and page information when available. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 102 repeats the phrase chunk metadata to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 03 - Fixed Windows

This pdf-playbook note describes fixed chunking uses bounded character windows with overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 003 repeats the phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes fixed chunking uses bounded character windows with overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 103 repeats the phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 04 - Structural Boundaries

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 004 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 104 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 05 - Embedding Backend

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 005 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 105 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 06 - Vector Storage

This pdf-playbook note describes FAISS is used when installed and NumPy vectors remain available as a portable fallback. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 006 repeats the phrase vector storage to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes FAISS is used when installed and NumPy vectors remain available as a portable fallback. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 106 repeats the phrase vector storage to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 07 - Retrieval Metrics

This pdf-playbook note describes comparison uses hit at k, reciprocal rank, source accuracy, and section accuracy. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 007 repeats the phrase retrieval metrics to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes comparison uses hit at k, reciprocal rank, source accuracy, and section accuracy. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 107 repeats the phrase retrieval metrics to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 08 - Operational Checks

This pdf-playbook note describes the pipeline verifies document count, estimated pages, chunk counts, and metadata coverage. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 008 repeats the phrase operational checks to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes the pipeline verifies document count, estimated pages, chunk counts, and metadata coverage. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 108 repeats the phrase operational checks to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 09 - Ingestion Contract

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 009 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 109 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 10 - Chunk Metadata

This pdf-playbook note describes every chunk carries source, title, section, chunk_id, offsets, and page information when available. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 010 repeats the phrase chunk metadata to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes every chunk carries source, title, section, chunk_id, offsets, and page information when available. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 110 repeats the phrase chunk metadata to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 11 - Fixed Windows

This pdf-playbook note describes fixed chunking uses bounded character windows with

overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 011 repeats the phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes fixed chunking uses bounded character windows with overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 111 repeats the phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 12 - Structural Boundaries

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 012 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 112 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 13 - Embedding Backend

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that

indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 013 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 113 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 14 - Vector Storage

This pdf-playbook note describes FAISS is used when installed and NumPy vectors remain available as a portable fallback. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 014 repeats the phrase vector storage to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes FAISS is used when installed and NumPy vectors remain available as a portable fallback. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 114 repeats the phrase vector storage to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 15 - Retrieval Metrics

This pdf-playbook note describes comparison uses hit at k, reciprocal rank, source accuracy, and section accuracy. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the

relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 015 repeats the phrase retrieval metrics to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes comparison uses hit at k, reciprocal rank, source accuracy, and section accuracy. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 115 repeats the phrase retrieval metrics to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 16 - Operational Checks

This pdf-playbook note describes the pipeline verifies document count, estimated pages, chunk counts, and metadata coverage. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 016 repeats the phrase operational checks to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes the pipeline verifies document count, estimated pages, chunk counts, and metadata coverage. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 116 repeats the phrase operational checks to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 17 - Ingestion Contract

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section

visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 017 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 117 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 18 - Chunk Metadata

This pdf-playbook note describes every chunk carries source, title, section, `chunk_id`, offsets, and page information when available. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 018 repeats the phrase chunk metadata to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes every chunk carries source, title, section, `chunk_id`, offsets, and page information when available. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 118 repeats the phrase chunk metadata to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 19 - Fixed Windows

This pdf-playbook note describes fixed chunking uses bounded character windows with overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be

relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 019 repeats the phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes fixed chunking uses bounded character windows with overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 119 repeats the phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 20 - Structural Boundaries

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 020 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 120 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 21 - Embedding Backend

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is

too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 021 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 121 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 22 - Vector Storage

This pdf-playbook note describes FAISS is used when installed and NumPy vectors remain available as a portable fallback. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 022 repeats the phrase vector storage to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes FAISS is used when installed and NumPy vectors remain available as a portable fallback. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 122 repeats the phrase vector storage to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 23 - Retrieval Metrics

This pdf-playbook note describes comparison uses hit at k, reciprocal rank, source accuracy, and section accuracy. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The

preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 023 repeats the phrase retrieval metrics to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes comparison uses hit at k, reciprocal rank, source accuracy, and section accuracy. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 123 repeats the phrase retrieval metrics to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 24 - Operational Checks

This pdf-playbook note describes the pipeline verifies document count, estimated pages, chunk counts, and metadata coverage. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 024 repeats the phrase operational checks to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes the pipeline verifies document count, estimated pages, chunk counts, and metadata coverage. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 124 repeats the phrase operational checks to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 25 - Ingestion Contract

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a

later answer can cite the exact file or page. Topic marker 025 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes loaders preserve source paths, document titles, page numbers, and a stable kind field. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 125 repeats the phrase ingestion contract to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 26 - Chunk Metadata

This pdf-playbook note describes every chunk carries source, title, section, chunk_id, offsets, and page information when available. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 026 repeats the phrase chunk metadata to make evaluation queries deterministic without hiding the normal prose around it.

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Chapter 27 - Fixed Windows

This pdf-playbook note describes fixed chunking uses bounded character windows with overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 027 repeats the

phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes fixed chunking uses bounded character windows with overlap and whitespace-aware break points. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 127 repeats the phrase fixed windows to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 28 - Structural Boundaries

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 028 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

This pdf-playbook note describes structural chunking starts from headings, pages, files, classes, and functions. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 128 repeats the phrase structural boundaries to make evaluation queries deterministic without hiding the normal prose around it.

Chapter 29 - Embedding Backend

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 029 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the

normal prose around it.

This pdf-playbook note describes Ollama embeddings are requested through the local HTTP API and stored as normalized vectors. The section is intentionally verbose so that indexing tests have enough material to form realistic chunks. A retrieval system should keep the relationship between the user question, the original source, and the surrounding section visible. When a chunk crosses a boundary, the answer may still be relevant, but the metadata becomes less precise for review and citation. When a chunk is too small, it can lose setup, definitions, and examples that make the result useful to a reader. The preferred implementation records offsets, stable identifiers, and plain text so that a later answer can cite the exact file or page. Topic marker 129 repeats the phrase embedding backend to make evaluation queries deterministic without hiding the normal prose around it.